Two Dimensional Vehicle Model

1 Overview

The vehicle model is described by three complementary parts:

1. a *dynamics model*, which describes how the vehicle state evolves;
2. a *constraints model*, which defines admissible states and motion commands; and
3. a *capability model*, which describes what the vehicle can currently achieve.

The separation is deliberate. The dynamics model is what the simulation core integrates. The constraints model is a *declaration* of the region in which the dynamics model is claimed to be valid. The capability model is the component's external interface: it answers questions about achievable motion without requiring the dynamics to be integrated forward, and it is what guidance, motion planning, and mission objective management consume.

The model is first introduced without discrete operating modes. A second formulation then extends the model with two propulsion modes, nominal and boost.

2 Conventions

Frame. A planar projection of the standard North – East – Down frame is used. The coordinate p_x is measured towards north and p_y towards east, both in metres. The vertical axis is suppressed; the vehicle is assumed to operate at constant altitude.

Heading. The heading ψ is measured clockwise from north, that is, from the positive p_x axis towards the positive p_y axis. This is the conventional aviation sense and is retained in preference to the mathematical convention.

Turn rate. The turn rate $\omega = \dot{\psi}$ is positive for a right turn, corresponding to a positive bank angle φ.

Airspeed. The state v is *airspeed.* Wind enters the position kinematics only. Consequently ψ is an air relative heading, and the ground track differs from ψ whenever the wind is non zero. This distinction is retained because the aerodynamic model is a function of airspeed while any external observer of the vehicle measures ground referenced motion.

Units. SI base units throughout. Angles are in radians.

3 Baseline Vehicle Model

3.1 Dynamics model

The baseline continuous time dynamics are written as

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\theta}, \boldsymbol{\eta}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{w}. \tag{1}$$

The state is

$$\boldsymbol{x} = \begin{bmatrix} p_x & p_y & \psi & v & m \end{bmatrix}^T \in \mathbb{R}^5, \tag{2}$$

where p_x and p_y are position coordinates, ψ is heading, v is airspeed, and m is total vehicle mass.

The continuous motion command is

$$\boldsymbol{u} = \begin{bmatrix} T & \omega \end{bmatrix}^T \in \mathbb{R}^2, \tag{3}$$

where T is thrust and ω is turn rate.

The vehicle parameter vector is

$$\boldsymbol{\theta} = \begin{bmatrix} c_p & c_i & c_\ell & c_{\text{TSFC}} \end{bmatrix}^T, \tag{4}$$

where c_p is the parasite drag parameter, c_i is the induced drag parameter, c_ℓ is the maximum lift parameter, and c_{TSFC} is the thrust specific fuel consumption parameter.

The environmental parameter vector is

$$\boldsymbol{\eta} = \begin{bmatrix} g & \rho \end{bmatrix}^T, \tag{5}$$

where g is gravitational acceleration and ρ is air density.

3.1.1 Aerodynamic model

The drag model follows the conventional parabolic drag polar [1, 2],

$$C_D = C_{D_0} + kC_L^2, \qquad k = \frac{1}{\pi e AR}, \tag{6}$$

where C_D is total drag coefficient, C_{D_0} is zero lift drag coefficient, C_L is lift coefficient, AR is wing aspect ratio, and e is the Oswald efficiency factor.

Dynamic pressure is $\frac{1}{2}\rho v^2$, and lift and drag are written as

$$L = \tfrac{1}{2}\rho v^2 S C_L, \qquad D = \tfrac{1}{2}\rho v^2 S C_D, \tag{7}$$

where S is the wing reference area [3].

Parasite drag. The parasite drag is

$$D_p = \tfrac{1}{2}\rho S C_{D_0} v^2. \tag{8}$$

Defining

$$c_p := \tfrac{1}{2} S C_{D_0}, \tag{9}$$

gives

$$D_p = \rho c_p v^2. \tag{10}$$

Induced drag. For induced drag, substitution of C_L from eq. (7) gives

$$D_i = \frac{2kL^2}{\rho S v^2}. \tag{11}$$

In straight, steady, level flight $L = mg$, giving

$$D_{i,0} = \frac{2k(mg)^2}{\rho S v^2}. \tag{12}$$

Using $k = 1/(\pi e AR)$ and defining

$$c_i := \frac{2g^2}{\pi e AR\, S}, \tag{13}$$

the baseline induced drag becomes

$$D_{i,0} = \frac{c_i}{\rho}\frac{m^2}{v^2}. \tag{14}$$

Maximum lift. The largest lift the vehicle can generate at a given airspeed follows from eq. (7) evaluated at the maximum lift coefficient $C_{L,\max}$. Defining

$$c_\ell := \tfrac{1}{2} S C_{L,\max}, \tag{15}$$

in direct analogy with eq. (9), gives

$$L_{\max}(v) = \rho c_\ell v^2. \tag{16}$$

Remark 1. *The three lumped parameters c_p, c_i and c_ℓ absorb the geometry S, C_{D_0}, e, AR and $C_{L,\max}$, so that a vehicle may be specified either by the lumped parameters directly or by the underlying geometry.*

3.1.2 Coordinated turn

During a coordinated level turn with bank angle φ, the vertical component of lift balances weight,

$$L\cos\varphi = mg, \tag{17}$$

so the load factor is

$$n = \frac{L}{mg} = \frac{1}{\cos\varphi}. \tag{18}$$

The horizontal lift component provides the centripetal force,

$$L\sin\varphi = \frac{mv^2}{R}, \tag{19}$$

where R is turn radius. Since $\omega = v/R$,

$$\omega = \frac{g\tan\varphi}{v}, \tag{20}$$

and consequently

$$n^2 = 1 + \left(\frac{v\omega}{g}\right)^2. \tag{21}$$

The sign convention of section 2 is consistent with eq. (20): a positive bank angle produces a positive turn rate and hence a right turn.

3.1.3 Total drag

Because induced drag is proportional to L^2, it scales with n^2. Using eq. (21), the total drag is

$$D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}) = \rho c_p v^2 + \frac{c_i}{\rho} \frac{m^2}{v^2} \left[1 + \left(\frac{v\omega}{g}\right)^2\right]. \tag{22}$$

It is convenient to name the induced drag coefficient at unit load factor,

$$A(v, m) := \frac{c_i m^2}{\rho v^2}, \tag{23}$$

so that eq. (22) may be written compactly as

$$D = \rho c_p v^2 + A(v, m) \left[1 + \left(\frac{v\omega}{g}\right)^2\right]. \tag{24}$$

3.1.4 Nominal dynamics

The nominal dynamics function is

$$\boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\theta}, \boldsymbol{\eta}) = \begin{bmatrix} v\cos\psi \\ v\sin\psi \\ \omega \\ \dfrac{T - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta})}{m} \\ -c_{\text{TSFC}} T \end{bmatrix}. \tag{25}$$

Remark 2. *Under the convention of section 2, with p_x towards north and ψ clockwise from north, the position kinematics retain the same algebraic form as under the mathematical convention. Only the interpretation of the axes and the sign of ω differ.*

A compact disturbance vector is

$$\boldsymbol{w} = \begin{bmatrix} w_x & w_y & w_\psi & F_v \end{bmatrix}^T, \tag{26}$$

where w_x and w_y are wind velocity components, w_ψ is a turn rate disturbance, and F_v is a signed longitudinal disturbance force.

The disturbance matrix is

$$\boldsymbol{G}(\boldsymbol{x}) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \frac{1}{m} \\ 0 & 0 & 0 & 0 \end{bmatrix}. \tag{27}$$

The complete baseline dynamics are therefore

$$\dot{p}_x = v\cos\psi + w_x, \tag{28}$$

$$\dot{p}_y = v\sin\psi + w_y, \tag{29}$$

$$\dot{\psi} = \omega + w_\psi, \tag{30}$$

$$\dot{v} = \frac{T - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}) + F_v}{m}, \tag{31}$$

$$\dot{m} = -c_{\text{TSFC}} T. \tag{32}$$

3.2 Constraints model

The baseline constraint parameters are collected in

$$\boldsymbol{\lambda} = \begin{bmatrix} T_{\min} & T_{\max} & n_{\max} & \omega_{\text{cap}} & v_{\min} & v_{\max} & m_{\text{dry}} \end{bmatrix}^T, \tag{33}$$

where $n_{\max}$ is the structural load factor limit and ω_{cap} is an absolute turn rate bound representing roll and control authority.

3.2.1 Turn rate limit

The turn rate is not bounded by a constant. Two distinct limits act, and the binding one depends on the operating point.

The *lift limited* load factor follows from eq. (16) and eq. (18),

$$n_\ell(v, m) = \frac{L_{\max}(v)}{mg} = \frac{\rho c_\ell v^2}{mg}, \tag{34}$$

and the available load factor is the smaller of the lift limit and the structural limit,

$$n_{\text{av}}(v, m) = \min\{\, n_{\max},\ n_\ell(v, m)\,\}. \tag{35}$$

Inverting eq. (21) gives the corresponding turn rate bound,

$$\omega_{\max}(v, m) = \begin{cases} \min\left\{\omega_{\text{cap}},\ \dfrac{g}{v}\sqrt{n_{\text{av}}^2(v, m) - 1}\right\}, & n_{\text{av}}(v, m) > 1, \\ 0, & \text{otherwise.} \end{cases} \tag{36}$$

Remark 3. *The lift limit dominates at low airspeed and the structural limit at high airspeed. The crossover, where $n_\ell(v, m) = n_{\max}$, is the corner speed defined in eq. (49), at which the instantaneous turn rate attains its maximum. A constant turn rate bound would imply an unbounded load factor as airspeed increases and would permit turns the wing cannot support at low airspeed.*

3.2.2 Admissible sets

The admissible input set is

$$\mathcal{U}(\boldsymbol{x}, \boldsymbol{\lambda}) = \left\{\boldsymbol{u} \in \mathbb{R}^2 : T_{\min} \le T \le T_{\max},\ |\omega| \le \omega_{\max}(v, m)\right\}. \tag{37}$$

The admissible state set is

$$\mathcal{X}(\boldsymbol{\lambda}) = \left\{\boldsymbol{x} \in \mathbb{R}^5 : v_{\text{s}}(m, 1) \le v \le v_{\max},\ v \ge v_{\min},\ m \ge m_{\text{dry}}\right\}, \tag{38}$$

where v_{s} is the stall speed defined in eq. (48).

Remark 4 (Enforcement is delegated)**.** *The constraints model* declares *the region in which the dynamics model is claimed to be valid. It does not specify what occurs when a command or a state lies outside that region, and the vehicle model performs no projection, saturation or clamping of its own.*

Responsibility for respecting $\mathcal{U}$ and $\mathcal{X}$ lies with the guidance and motion planning logic, possibly in combination with a runtime assurance mechanism interposed between the planner and the vehicle. Placing enforcement in the vehicle would conceal control design faults, would make the enforcement policy an unstated property of the vehicle rather than an explicit design decision, and would prevent different enforcement strategies from being compared against the same vehicle.

Outside $\mathcal{U}$ and $\mathcal{X}$ the dynamics remain integrable but the model is not claimed to be a faithful representation of the vehicle.

Remark 5. *The input set $\mathcal{U}$ is state dependent, since $\omega_{\max}$ is a function of airspeed and mass. The vehicle exposes $\omega_{\max}(v, m)$ through the capability model, so that guidance logic may evaluate the bound at the current operating point without reproducing the aerodynamic model.*

3.3 Capability model

The capability model is written generally as

$$\boldsymbol{c} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{\theta}, \boldsymbol{\eta}, \boldsymbol{\lambda}, \boldsymbol{w}). \tag{39}$$

A capability vector, evaluated at a specified turn rate ω, is

$$\boldsymbol{c} = \begin{bmatrix} T_{\mathrm{av}} & T_{\mathrm{req}} & a_{\max} & a_{\min} & n_{\mathrm{av}} & \omega_{\mathrm{av}} & \omega_{\mathrm{sus}} & R_{\min} & v_{\mathrm{s}} & v_{\mathrm{c}} & m_{\mathrm{fuel}} & t_{\mathrm{end}} & r_{\mathrm{sp}} \end{bmatrix}^T. \tag{40}$$

Thrust. The available thrust is

$$T_{\mathrm{av}} = \begin{cases} T_{\max}, & m > m_{\mathrm{dry}}, \\ 0, & m \le m_{\mathrm{dry}}. \end{cases} \tag{41}$$

The thrust required to hold the current airspeed at the specified turn rate is

$$T_{\mathrm{req}}(v, m, \omega) = D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}). \tag{42}$$

Longitudinal acceleration. For a specified turn rate ω, the instantaneous maximum and minimum longitudinal accelerations are

$$a_{\max} = \frac{T_{\mathrm{av}} - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}) + F_v}{m}, \tag{43}$$

$$a_{\min} = \frac{T_{\min} - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}) + F_v}{m}. \tag{44}$$

Instantaneous turn performance. The available load factor n_{av} is given by eq. (35) and the available turn rate by

$$\omega_{\mathrm{av}}(v, m) = \omega_{\max}(v, m), \tag{45}$$

with the corresponding minimum turn radius

$$R_{\min} = \frac{v}{\omega_{\mathrm{av}}}, \qquad \omega_{\mathrm{av}} > 0. \tag{46}$$

Sustained turn performance. The available turn rate is instantaneous: holding it in general requires drag to exceed available thrust, so airspeed decays. The *sustained* turn rate is the largest turn rate at which thrust and drag balance, so that $\dot{v} = 0$ in the absence of disturbance. Setting $T_{\mathrm{av}} = D(v, m, \omega)$ in eq. (24) and solving for ω gives the closed form

$$\omega_{\mathrm{sus}}(v, m) = \min\left\{ \omega_{\mathrm{av}}(v, m),\ \frac{g}{v}\sqrt{\frac{T_{\mathrm{av}} - \rho c_p v^2 - A(v, m)}{A(v, m)}} \right\}, \tag{47}$$

defined whenever $T_{\mathrm{av}} > \rho c_p v^2 + A(v, m)$, and zero otherwise. The condition for a non zero sustained turn rate is precisely the condition that the vehicle can maintain wings level flight at the given airspeed.

Remark 6. *The interval $[\omega_{\mathrm{sus}}, \omega_{\mathrm{av}}]$ is the region in which the vehicle may turn only by expending kinetic energy. The width of this interval is the quantity a motion planner trades against, and is therefore exposed explicitly rather than left to be inferred.*

Characteristic speeds. The stall speed at load factor n follows from inverting eq. (34),

$$v_{\mathrm{s}}(m,n)=\sqrt{\frac{nmg}{\rho c_{\ell}}}, \tag{48}$$

and the corner speed, at which the lift and structural limits coincide and the instantaneous turn rate is maximised, is

$$v_{\mathrm{c}}(m)=v_{\mathrm{s}}(m,1)\sqrt{n_{\max}}=\sqrt{\frac{n_{\max}mg}{\rho c_{\ell}}}. \tag{49}$$

Fuel and endurance. Remaining fuel mass is

$$m_{\mathrm{fuel}}=\max\{m-m_{\mathrm{dry}},\,0\}. \tag{50}$$

At the thrust required for steady flight, an instantaneous endurance estimate is

$$t_{\mathrm{end}}=\frac{m_{\mathrm{fuel}}}{c_{\mathrm{TSFC}}T_{\mathrm{req}}}, \qquad T_{\mathrm{req}}>0, \tag{51}$$

and the corresponding specific range, that is, distance flown per unit mass of fuel, is

$$r_{\mathrm{sp}}=\frac{v}{c_{\mathrm{TSFC}}T_{\mathrm{req}}}. \tag{52}$$

Remark 7. *The speed for maximum endurance minimises* $T_{\mathrm{req}}(v,m,0)$ *and the speed for maximum range maximises* $v/T_{\mathrm{req}}(v,m,0)$*. Both follow from eq.* (22) *by differentiation and are available to mission objective management as derived quantities.*

Remark 8 (True and believed capability)**.** *Evaluating eq.* (39) *requires the disturbance* $\boldsymbol{w}$*, which is a property of the environment and is not observable by the vehicle. Two distinct evaluations are therefore meaningful:*

$$\boldsymbol{c}=\boldsymbol{g}(\boldsymbol{x},\boldsymbol{\theta},\boldsymbol{\eta},\boldsymbol{\lambda},\boldsymbol{w}), \qquad \hat{\boldsymbol{c}}=\boldsymbol{g}(\hat{\boldsymbol{x}},\boldsymbol{\theta},\boldsymbol{\eta},\boldsymbol{\lambda},\hat{\boldsymbol{w}}), \tag{53}$$

where $\hat{\boldsymbol{x}}$ *and* $\hat{\boldsymbol{w}}$ *denote estimates. The former is the true capability and is available only to the simulation. The latter is the capability the vehicle believes it has, and is what guidance and planning must use. The discrepancy between the two is a property of the estimation problem and is intentionally preserved rather than removed.*

4 Vehicle Model with Nominal and Boost Modes

4.1 Mode dependent dynamics

The vehicle is extended with the discrete propulsion mode

$$q\in\mathcal{Q}, \qquad \mathcal{Q}=\{\mathrm{nom},\,\mathrm{boost}\}. \tag{54}$$

The continuous motion command is unchanged. The continuous state is augmented with a scalar *thermal state* s, introduced in section 4.3, which accumulates while boost is engaged:

$$\boldsymbol{x}=\begin{bmatrix}p_x & p_y & \psi & v & m & s\end{bmatrix}^T, \qquad \boldsymbol{u}=\begin{bmatrix}T & \omega\end{bmatrix}^T. \tag{55}$$

The switched dynamics are written as

$$\dot{\boldsymbol{x}}=\boldsymbol{f}_q(\boldsymbol{x},\boldsymbol{u},\boldsymbol{\theta},\boldsymbol{\eta})+\boldsymbol{G}(\boldsymbol{x})\boldsymbol{w}, \qquad q\in\mathcal{Q}. \tag{56}$$

The parameter vector becomes

$$\boldsymbol{\theta} = \begin{bmatrix} c_p & c_i & c_\ell & c_{\text{TSFC}}^{\text{nom}} & c_{\text{TSFC}}^{\text{boost}} & \tau_{\text{h}} & \tau_{\text{c}} \end{bmatrix}^T, \tag{57}$$

with

$$c_{\text{TSFC}}^{\text{boost}} > c_{\text{TSFC}}^{\text{nom}}, \tag{58}$$

where τ_{h} and τ_{c} are the thermal accumulation and recovery time constants.

The mode dependent fuel consumption parameter is

$$c_{\text{TSFC},q} = \begin{cases} c_{\text{TSFC}}^{\text{nom}}, & q = \text{nom}, \\ c_{\text{TSFC}}^{\text{boost}}, & q = \text{boost}, \end{cases} \tag{59}$$

and the mode dependent dynamics are

$$\boldsymbol{f}_q = \begin{bmatrix} v\cos\psi \\ v\sin\psi \\ \omega \\ \dfrac{T - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta})}{m} \\ -c_{\text{TSFC},q} T \\ \sigma_q(s) \end{bmatrix}, \qquad \sigma_q(s) = \begin{cases} \dfrac{1}{\tau_{\text{h}}}, & q = \text{boost}, \\ -\dfrac{s}{\tau_{\text{c}}}, & q = \text{nom}. \end{cases} \tag{60}$$

The drag model and the disturbance model are unchanged. Boost mode increases fuel consumption per unit thrust and, through the constraints model, permits a larger thrust command.

4.2 Mode dependent constraints model

The constraint parameters are extended with mode dependent propulsion limits:

$$\boldsymbol{\lambda} = \begin{bmatrix} T_{\min} & T_{\max}^{\text{nom}} & T_{\max}^{\text{boost}} & n_{\max} & \omega_{\text{cap}} & v_{\min} & v_{\max}^{\text{nom}} & v_{\max}^{\text{boost}} & m_{\text{dry}} & m_{\text{res}} & s_{\max} \end{bmatrix}^T, \tag{61}$$

with

$$T_{\max}^{\text{boost}} > T_{\max}^{\text{nom}}, \qquad v_{\max}^{\text{boost}} > v_{\max}^{\text{nom}}, \tag{62}$$

where m_{res} is a fuel reserve below which boost is inhibited and $s_{\max}$ is the thermal limit.

The mode dependent input set is

$$\mathcal{U}_q(\boldsymbol{x}, \boldsymbol{\lambda}) = \{\boldsymbol{u} : T_{\min} \le T \le T_{\max}^q,\ |\omega| \le \omega_{\max}(v, m)\}, \tag{63}$$

and the mode dependent state set is

$$\mathcal{X}_q(\boldsymbol{\lambda}) = \{\boldsymbol{x} : v_{\text{s}}(m, 1) \le v \le v_{\max}^q,\ v \ge v_{\min},\ m \ge m_{\text{dry}},\ 0 \le s \le s_{\max}\}. \tag{64}$$

Remark 9 (The turn rate bound is mode independent)**.** *The bound $\omega_{\max}(v, m)$ of eq.* (36) *does not carry a mode index. The instantaneous turn rate is limited by available lift and by structural strength, neither of which is affected by propulsion setting. An earlier formulation in which $\omega_{\max}^{\text{boost}} > \omega_{\max}^{\text{nom}}$ is not consistent with the aerodynamic model.*

Boost nonetheless improves turn performance, but it does so through the sustained *turn rate eq.* (47)*, which depends on available thrust and is therefore mode dependent. This is the physically correct channel: boost does not allow a tighter turn, it allows a given turn to be held without losing airspeed.*

4.3 Mode switching constraints

The mode is a discrete control input, but transitions between modes are not assumed to be unrestricted. Let q denote the current mode and q^+ the mode after a switching decision. Admissible transitions are described by

$$q^+ \in \mathcal{S}_q(\boldsymbol{x}, \boldsymbol{\lambda}) \subseteq \mathcal{Q}. \tag{65}$$

A concrete instantiation sufficient for the present model is

$$\mathcal{S}_q(\boldsymbol{x}, \boldsymbol{\lambda}) = \begin{cases} \{\text{nom}\}, & s \geq s_{\max} \ \text{ or } \ m \leq m_{\text{dry}} + m_{\text{res}}, \\ \{q\}, & t - t_q < \Delta t_{\text{dwell}}, \\ \{\text{nom}, \text{boost}\}, & \text{otherwise}, \end{cases} \tag{66}$$

where t_q is the time of the most recent mode transition and Δt_{dwell} is a minimum dwell time. The cases are evaluated in the order given.

Remark 10 (Being forced out and being locked in are different)**.** *An earlier formulation of* eq. (66) *listed the two permissive cases by current mode and returned* $\{\text{nom}\}$ *otherwise. That conflates two situations which happen to share a branch. Exhausting the thermal or fuel budget must force the aircraft* out *of boost; not having served the dwell must* keep *it where it is. Under the earlier formulation,* $q =$ boost *with the dwell unserved returned* $\{\text{nom}\}$*, so boost was granted on one step and revoked on the next indefinitely — an anti-chattering condition that produced chattering. It was found by simulating the model rather than by reading it, and the thermal state never rose above two per cent of its limit.*

The ordering is also load-bearing. Were the dwell tested first, an aircraft reaching $s_{\max}$ *shortly after engaging would be held in boost past its own thermal limit, and the guard against chattering would defeat the guard against destroying the engine.*

Three distinct effects are captured. The thermal state s, evolving according to eq. (60), limits the duration of continuous boost and imposes a recovery period afterwards. The fuel reserve m_{res} prevents boost from being selected when the remaining fuel is committed to recovery. The dwell time Δt_{dwell} prevents chattering, which is otherwise a hazard both for the planner and for numerical integration of the switched system.

Only the first of the three is physical. An engine genuinely cannot sustain boost past its thermal limit, whereas an aircraft can light its afterburner on the last hundred kilograms of fuel and can switch modes twice in a second; it simply should not. The fuel reserve and the dwell time are therefore declared policy, carried in $\boldsymbol{\lambda}$ alongside $v_{\min}$, which is likewise a declared floor rather than an aerodynamic one. The distinction matters to anyone configuring a different aircraft: the thermal limit is the entry they cannot simply relax.

Remark 11 (The switching set is declared, not enforced)**.** *The mode is a discrete input, so* eq. (65) *is an input constraint, and the declaration/enforcement separation of the constraints model applies to it unchanged: the vehicle* declares $\mathcal{S}_q$ *and does not enforce it. A projection is offered, and because a mode cannot be clipped by degree the way a thrust command can, that projection is a fallback to* nom *rather than a partial reduction. A commanded mode outside* $\mathcal{S}_q$ *that is flown anyway remains integrable, is not claimed valid, and shows itself through* $\mathcal{X}_q$*:* s *runs past* $s_{\max}$ *and is reported. This is the same treatment a thrust command above* $T^q_{\max}$ *receives.*

Remark 12 (The dwell condition is an argument, not a state)**.** $t - t_q$ *is history rather than state, so a declaration written as a pure function of* $\boldsymbol{x}$ *and* $\boldsymbol{\lambda}$ *cannot evaluate it. Two resolutions are available: carry* t_q *as an additional state, or supply the elapsed time to the declaration as* *an argument. The latter is preferred here. Adding a state to the vector would place a decision-hygiene quantity inside the integrator and the Jacobian, where it participates in nothing, and the party that selects the mode necessarily observes its own transitions and can supply the interval at no cost.*

Remark 13. *Equation* (66) *is one instantiation of the abstract relation eq.* (65) *and is not intended to be exhaustive. Further restrictions related to engine condition, operating point, or externally imposed rules of engagement may be introduced without altering the structure of the model, provided they are expressible as a function of* $\boldsymbol{x}$ *and* $\boldsymbol{\lambda}$.

4.4 Mode dependent capability model

The capability model becomes

$$\boldsymbol{c} = \boldsymbol{g}_q(\boldsymbol{x}, \boldsymbol{\theta}, \boldsymbol{\eta}, \boldsymbol{\lambda}, \boldsymbol{w}), \tag{67}$$

with the following quantities acquiring a mode index.

The currently available thrust is

$$T_{\mathrm{av},q} = \begin{cases} T_{\max}^{\mathrm{nom}}, & q = \mathrm{nom},\ m > m_{\mathrm{dry}}, \\ T_{\max}^{\mathrm{boost}}, & q = \mathrm{boost},\ m > m_{\mathrm{dry}}, \\ 0, & m \le m_{\mathrm{dry}}. \end{cases} \tag{68}$$

The maximum longitudinal acceleration follows as

$$a_{\max,q} = \frac{T_{\mathrm{av},q} - D(v, m, \omega; \boldsymbol{\theta}, \boldsymbol{\eta}) + F_v}{m}, \tag{69}$$

and the sustained turn rate from eq. (47) evaluated with $T_{\mathrm{av},q}$,

$$\omega_{\mathrm{sus},q}(v, m) = \min\left\{ \omega_{\mathrm{av}}(v, m),\ \frac{g}{v}\sqrt{\frac{T_{\mathrm{av},q} - \rho c_p v^2 - A(v, m)}{A(v, m)}} \right\}. \tag{70}$$

By remark 9, the instantaneous quantities n_{av}, ω_{av}, $R_{\min}$, v_{s} and v_{c} are mode independent.

The endurance estimate becomes

$$t_{\mathrm{end},q} = \frac{m_{\mathrm{fuel}}}{c_{\mathrm{TSFC},q} T_{\mathrm{req}}}, \qquad T_{\mathrm{req}} > 0. \tag{71}$$

The capability model also exposes whether boost is currently selectable, and for how long it could be held,

$$c_{\mathrm{boost}} = \begin{cases} 1, & \mathrm{boost} \in \mathcal{S}_q(\boldsymbol{x}, \boldsymbol{\lambda}), \\ 0, & \mathrm{boost} \notin \mathcal{S}_q(\boldsymbol{x}, \boldsymbol{\lambda}), \end{cases} \qquad t_{\mathrm{boost}} = \tau_{\mathrm{h}} \left(s_{\max} - s \right), \tag{72}$$

where t_{boost} is the time remaining before the thermal limit is reached at continuous boost. Both quantities are required by mission objective management, which must reason about whether a boost committed now remains available later.

5 Compact Summary

The baseline model is

$$\begin{aligned} &\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\theta}, \boldsymbol{\eta}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{w}, \\ &\boldsymbol{x} \in \mathcal{X}(\boldsymbol{\lambda}), \\ &\boldsymbol{u} \in \mathcal{U}(\boldsymbol{x}, \boldsymbol{\lambda}), \\ &\boldsymbol{c} = \boldsymbol{g}(\boldsymbol{x}, \boldsymbol{\theta}, \boldsymbol{\eta}, \boldsymbol{\lambda}, \boldsymbol{w}). \end{aligned} \tag{73}$$

The mode dependent extension is

$$
\begin{aligned}
\dot{\boldsymbol{x}} &= \boldsymbol{f}_q(\boldsymbol{x}, \boldsymbol{u}, \boldsymbol{\theta}, \boldsymbol{\eta}) + \boldsymbol{G}(\boldsymbol{x})\boldsymbol{w}, \qquad q \in \mathcal{Q}, \\
\boldsymbol{x} &\in \mathcal{X}_q(\boldsymbol{\lambda}), \\
\boldsymbol{u} &\in \mathcal{U}_q(\boldsymbol{x}, \boldsymbol{\lambda}), \\
q^+ &\in \mathcal{S}_q(\boldsymbol{x}, \boldsymbol{\lambda}), \\
\boldsymbol{c} &= \boldsymbol{g}_q(\boldsymbol{x}, \boldsymbol{\theta}, \boldsymbol{\eta}, \boldsymbol{\lambda}, \boldsymbol{w}).
\end{aligned} \tag{74}
$$

In both cases the membership relations are declarations of validity rather than enforced projections, in the sense of remark 4.

6 Assumptions and Limitations

The model assumes a planar point mass vehicle, a coordinated steady level turn, negligible sideslip, negligible altitude change, negligible roll transients, instantaneous adjustment of lift, and validity of the parabolic drag polar within the considered operating region. The coordinated turn relations are standard results in aircraft flight mechanics [4].

Several consequences follow from the planar assumption and are recorded explicitly:

1. Potential energy is absent. Manoeuvres that trade altitude for airspeed, or the converse, cannot be represented, and the energy state of the vehicle is fully described by airspeed.

2. Air density ρ is a fixed environmental parameter rather than a function of altitude, so the model describes a single flight level.

3. Thrust specific fuel consumption is constant within a mode, whereas in practice it depends on airspeed and altitude.

4. Turn rate and thrust are commanded directly, so roll response and engine spool dynamics are absent. A first order lag on either command, $\dot{\omega} = (\omega_{\mathrm{c}} - \omega)/\tau_\omega$ and $\dot{T} = (T_{\mathrm{c}} - T)/\tau_T$, may be introduced by augmenting the state without altering the remainder of the model.

5. The induced drag model degrades as airspeed approaches the stall speed eq. (48), and the model is not claimed to be valid below it.

References

[1] NASA Glenn Research Center, *Drag Coefficient*, Beginner's Guide to Aeronautics. `https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/drag-coefficient/`

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